Expected Goals Modelling in The Premier League

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Introduction

The outcome of a football match can be identified by the scoreline, a win is a win regardless of the underlying performance. By using basic summary statistics, a more detailed understanding of performance can be achieved. For example, a team that had more shots than their opponents may appear to be the stronger team because shots are an attacking action and attacking actions lead to goals and therefore wins. However, using raw totals of statistics does not represent the 'story' of the game, a team with lots of shots can still lose to a team with only a few. A variety of advanced metrics have been developed to better 'tell the story' of a match and help fans understand why the team with lots of shots might have lost.

Expected Goals (xG) is a mainstream advanced metric used across the sport. It is an intricate measure of shooting performance that can be represented by just one number, making it easily accessible to those without a deeper understanding of statistics or data analytics. Early literature identifies that xG positions itself well as a 'comparative tool for team or player evaluation purposes' (Ensum, Pollard and Taylor, 2004) and that it is well suited to 'predicting players and teams goal scoring probabilities' (Rathke, 2017). In the present, it is most often used as a discussion point after watching a match and is used by broadcast companies such as Sky or BBC to 'analyse' a match or by clubs to evaluate individual and team performance.

This report is based on the associated xG model produced for the assignment, the remainder of this introduction is a literature review that aims to discuss the influence of key authors on the topic and how they informed the choices made when developing the model.

xG is a measure of a player's 'probability of scoring from any given shot' (Ensum, Pollard and Taylor, 2004) and is made up of a series of predictors such as distance the shot travelled or which foot a player used. The most prevalent predictors in the literature are the distance to the goal and the angle of the goal available to the shooter (Ensum, Pollard and Taylor, 2004), (Rathke, 2017), (Mead, O'Hare and McMenemy, 2023). These predictors are constructed based on the coordinates of a shot and the goal, which are widely available in football data so were identified as key predictors for the model.

As well as spatial data of the shot position, many additional features are considered to have an impact on xG. Lucey et al. (2015) showed that considering the context of a shot has influence on the probability of scoring, 'counter attacking' (fast-break), 'open-play' (regular-play), 'corners' and 'free-kicks' have all been included in the model as additional context features. Mead, O'Hare and McMenemy (2023) identified that positional groups may also influence the probability, with forwards having the 'highest odds of scoring' within betting markets, so players were split into positional groups for the model.

Across the literature, logistic regression is the standard method of determining xG (Ensum, Pollard and Taylor, 2004), (Rathke, 2017), (Mead, O'Hare and McMenemy, 2023). A number of model families are discussed by Mead, O'Hare and McMenemy (2023), such as decision trees, AdaBoost and XGBoost. This paper was key for deciding which model families to choose. A generalised linear model was chosen as a relatively simple baseline used to determine if the more complex and tuneable models, such as random forest and XGBoost, significantly improved the overall model. The performance metric of choice for the model was also determined by the literature. Anzer and Bauer (2021) uses 'area under the curve' (AUC) as a form of model evaluation but deem alternative metrics to be more significant due to AUC only measuring the success of binary outcomes rather than a probability. Therefore, the model uses log-loss in accordance with Mead, O'Hare and McMenemy (2023) due to its ability to evaluate a model's ability to 'accurately estimate goal probability' rather than just if the shot resulted in a goal or not.

Data and Feature Construction C.1 - 3

The original data contained a variety of variables, including some that had no influence on shot data. Section C.1 of the code begins with some initial inspections of the data, such as checking what actions are considered a shot (SavedShot, Goal, MissedShots and ShotOnPost) and which columns interact with shots. Moving onto section C.2, this is where the shot location coordinates and contextual features were identified. Positional groups were also assigned at this stage. In section C.3, the target variable 'goal' was initiated and transformed into a factor, it is important that the target variable is considered a categorical outcome rather than a continuous outcome.

As per the literature discussed and the data available, the key predictor variables/features chosen for this model were:

- Distance
- Header
- Left Foot
- Fast Break
- Corner
- Regular Play
- Position Group
- Angle
- Right Foot
- Big Chance
- Direct Freekick
- Assisted
- Minute

All the variables used, except distance, angle, minute and position group, are contextual features represented as binary indicators 0 for no and 1 for yes. The position groups are categoric and split up into GK, DEF, MID, ATT based on the shot taker's playing position.

For example, if a shot was taken with a ST's left foot on a counterattack and was assisted by another player but they did not score, the data point would look like:

	goal	distance	angle	header	right_foot	left_foot	big_chance	fast_break	direct_freekick	from_corner	assisted	regular_play	minute	position_group
1	0	13.2177	23.89885	0	0	1	0	1	0	0	1	0	92	ATT

Figure 1: Example Data Point

Of those listed, three features had to be constructed using the data available. Distance was constructed using the shot x and y coordinates and the goal x and y coordinates. The

Euclidean Distance is used as it is the shortest possible straight path from the shot to the centre of the goal. Angle towards the goal (available to the shot-taker) was constructed using the coordinates of the shot and both goalposts, a more central shot close to the goal should have a larger angle available and as the shot position moves wider the angle available should decrease. This means that a shot that converted to a goal should, on average, have a lower distance and a larger angle, as can be seen in the summary table below.

	Mean Distance	Mean Angle	Number of shots
No Goal	18.45	24.00	8739
Goal	12.26	40.07	979

Table 1: Summary table of distance and angle

Alongside the summary table, a shot map (figure 2) was produced using the coordinates as a precautionary check that the location of the shots were reasonable. According to the figure, most shots were in the attacking (attacking left to right) half of the field and concentrated within the opposition penalty area.

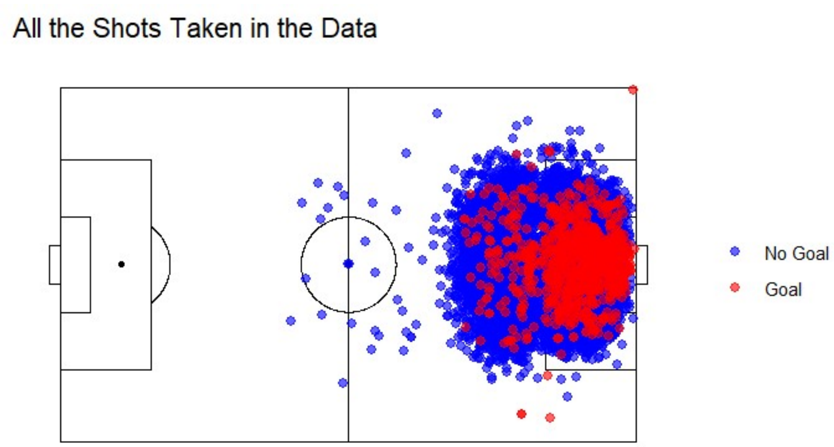

Figure 2: Shot Map of All Shots

Modelling C.4 - 11

The data is initially split into a train-test division with a ratio of 80-20, where the outcome variable is stratified to ensure a reasonable ratio of goals to not goals in each split. This is an important step because there are a lot more shots that do not result in a goal than do, so keeping a reasonable ratio ensures more reliability. The 20% test set remains untouched until the final predictions are ready to be made, and the model is ready for overall evaluation. The training data is used to create five cross-validation folds, again with goal stratified across them. These folds are vital for tuning grids later in the modelling process as a form of resampling that allows the test data set to remain completely sealed to help prevent data leakage. The test set is only used as a final evaluation of the models. The next section discusses preprocessing steps; all of these recipes are only based on the training data to prevent data leakage.

As per the code in section C.5, three recipes are constructed with the appropriate preprocessing steps implemented. First, a recipe for a generalised linear model (GLMNet) with preprocessing steps of step_dummy to produce dummy binary-indicator variables for the categorical position groups, step_normalize to ensure the penalty is in line with the scaled numeric predictors, and step_lincomb to remove any columns with linear combinations, although not required it was added as a precaution. Second, a recipe for generic tree models (decision tree and random forest) that only required step_lincomb as a precaution due to tree models' ability to deal with categorical variables without any preprocessing. Last, the recipe for the more complex tree model (XGBoost), using step_lincomb in the same way as the previous two but including step_dummy was vital as the tidymodels XGBoost engine requires numeric/binary variables rather than categoric.

Moving onto section C.6, which includes the model specifications. Five total models were included in the initial workflow; the exact specifications can be found in Table 2.

Table 2: Initial Model Specifications

Model	Engine	Mode
GLMNet	GLMNet	Classification
Decision Tree	Rpart	Classification
Random Forest	Ranger	Classification
Basic XGBoost	XGBoost	Classification
Advanced XGBoost	XGBoost	Classification

As previously mentioned, GLMNet was used as a reasonable baseline as it doesn't require much tuning and was represented well in the literature. Decision tree was included in case of a need for easy interpretation of the workflow but was not expected to perform to the same degree as GLMNet or XGBoost. Random forest was chosen because of its appearance in the literature and as a reasonable step between a linear model and a supervised learning model like XGBoost. Two different XGBoost specifications were chosen to inspect if a highly tuned model would perform significantly better than a slightly tuned model. These specifications were used as a first pass-through of the data, which was used to refine the tuning ranges as appropriate.

Section C.7 of the code depicts predictor mapping for the models that include mtry, random forest and XGBoost. It's a unique hyper-parameter that decides the number of predictors to consider at each split, so the maximum range is the number of predictors, 13. However due to the dummy variables produced in place of the position category leaves XGBoost with 15 predictors instead. The recipes were prepped using the training data only to avoid data leakage.

Moving onto the tuning parameters for each model, section C.8 is the set up for the first run-through. Each set of pre-initiated recipes and specifications are combined along with parameter ranges according to the specification. The ranges were mostly kept broad as the aim of this set of ranges was to capture the best direction of tuning and identify the best models rather than try and get the perfect model. For example, the GLMNet parameters were given a range of 0-1 for mixture, which is the complete range between ridge and lasso to allow the model to identify the correct mixture. As previously mentioned, random forest and XGBoost used the number of predictors as their maximum range for mtry, but the difference between random forest and the two XGBoost models was to try and make incremental steps up to a complex model. Random forest only requires a small amount of tuning and acts as a good baseline for the more complex models. The key difference between the two XGBoost models were the parameters that were tuned, the advanced model was allowed to explore a wider range of possibilities.

The five models were then combined into one workflow set in order to race them within the same workflow. Race_tune_anova allows the grid search to evaluate all the tuneable parameters by fitting the model with a different set of parameters each time across 4 of the 5 validation folds and then tested on the final fold. However, racing allows for a combination of parameters to be discarded as early as 3 folds, burn in = 3, if there is not an improvement to the evaluation metric. As mentioned, log loss was the chosen metric, the lower the value meaning that the model is better at predicting if a shot becomes a goal or not.

For the second run the number of tuning grids was reduced from 600 to 300, this can be done to decrease the time taken for tuning and can be done because the second run through is going through less parameters combinations as the tuning ranges have reduced. It is also viable because the second run is only looking to potentially improve on the best models rather than identify the best parameters from a larger range. From the initial run through, GLMNet, random forest and XGBoost performed the best according to their log loss results. GLMNet and the advanced XGBoost had a mean log loss of 0.262 but were separated by their standard error, 0.00360 and 0.00371 respectively. Random forest had a mean log loss of 0.264 so slightly worse but it did have a lower standard error of 0.00324. Decision Tree was the worst performer at 0.269 and the basic XGBoost had a mean of 0.267 so the decision to tune a more complex XGBoost paid off. Using the models that performed significantly better, the decision was made to run a second race with these models and a more refined tuning grid for each. The tuning ranges were updated according to the best combinations in the previous race, focusing the gird on the best ranked parameter combinations for each model means that fewer grids need to be used on the second run through. The goal of this race is to try and identify the best possible model from a selection of good models.

The advanced XGBoost model was split into two differently tuned models. Model A was optimised for speed and simplicity, another check to see if the extra tuning had a significant improvement on results. Model B was tuned to use more trees and a slower learn rate, this model is less likely to overfit due to the smaller corrections made by each tree (smaller learn rate). Of this set of results, the XGBoost_A model performed the best, but its log loss was tied with XGBoost_B and GLMNet at 0.262. As there was no significant improvement to the model’s evaluation metric, alternative measures must be considered such as speed, simplicity and data availability which will be discussed in the results section.

Results C.12 - 14

rank	wflow_id	model	mean	std_err
1	xgb_advanced_A_xgb_advanced_A	boost_tree	0.2617658	0.004190183
2	xgb_advanced_B_xgb_advanced_B	boost_tree	0.2619021	0.004209956
3	glmnet_glmnet	logistic_reg	0.2619841	0.003655401
4	random_forest_random_forest	rand_forest	0.2638500	0.003160722

Table 3: Evaluation Metrics Results of Models

This section begins with Table 3, showing the models used in the second run through. Select_best was used to rank the models according to their log loss performance on the cross-validation performance rather than their prediction results on the test set. The two XGBoost models performed the best with XGBoost_A slightly outperforming the more complex model, however the GLMNet model was within 0.00022 of the best performing models and had a significantly lower standard error. Random Forest did not perform to the same standards as the other three according to the evaluation metric, so it was removed from contention for the final model. As the top three models performed at a similar level, alternative measures had to be judged in order to choose a final model for predictions.

GLMNet was chosen as the final model due to its high performance on the evaluation metric and its overall simplicity, the complexity and extra tuning requirements of the XGBoost models did not significantly improve on the GLMNet model so model complexity and computational power were key factors considered when choosing the final model. The best combination of parameters according to the mean log loss was a penalty of 0.00449 and a mixture of 0.0208. A small penalty means there was little shrinkage, most of the predictor variables were kept. The mixture was closest to 0, which means it was ridge regression, and again suggests that the predictor variables have significant influence on the result as they are gradually shrunk toward zero rather than being removed completely as a lasso regression would. The final predictions on the test set gave a log loss of 0.273 so a slight increase from the training data, but that is to be expected. As the increase isn't substantial, the model does not appear to be subject to overfitting and performed reasonably well on the sealed test data. As the model performed well, it was used to predict xG results for the full data which included player names and shot coordinates which will be used for visualisations in this section.

Ridge regression does not reduce any coefficient's influence to zero, so there may be instances of multicollinearity which causes the sign (+/-) of a coefficient's impact to be flipped. This can be seen in Figure 3.

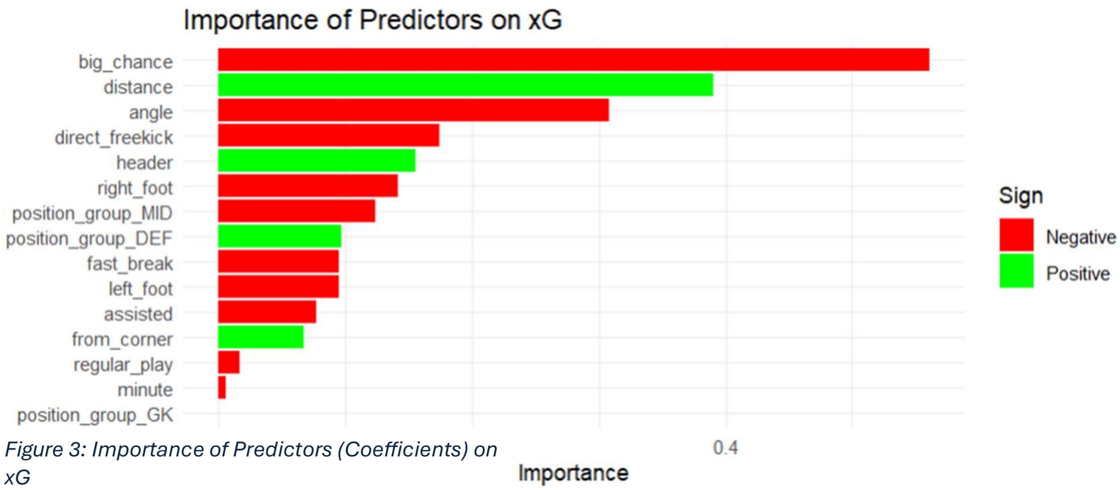

Figure 3: Importance of Predictors (Coefficients) on xG

Big_chance is shown to have a negative impact on xG, but in reality it is sharing influence with the other most influential predictors, distance and angle. Big_chance is a subjective tag given to shots, and often a shot is considered a big chance when it is a central shot close to the goal. It is best to only interpret the size of the bar in this style of plot rather than the sign due to the multicollinearity issue. There is a chance that this reaction is happening with direct_freekick as well, but in a footballing context a set piece gives the goalkeeper and defenders time to get in a good position to save/block the shot before the shot occurs, so they are better prepared and are more likely to stop the shot. Another concern of the model is that a headed shot shows to have a positive influence on xG, initially this is not what is expected as headed shots are often less powerful and easier to save. However, the consideration of shot location of a header compared to a normal shot is that a player would only attempt a header in very particular instances, often from a corner or a cross that meets the head of the player, usually centrally, in the box, so the distance to the goal is going to be relatively low and the angle of the goal available to the shot taker would be high, therefore a positive influence on xG is reasonable within the footballing context. Minute had very little influence on xG, this suggests that the model does not consider any extra pressure that a late, game-deciding shot has compared to a shot in regular time. Regular play and the goalkeeper position group also had little effect. On reflection big_chance would not be used in a model in the future due to its influence being a result of a subjective post-shot tag that may be significantly based on the position of the shot.

Figure 4 is a heatmap of all the predicted xG values, it shows that shots that are more central and closer to the goal have a higher xG (yellow/green/blue) which means the model is performing as expected. The plot suggests that shot location is the most influential predictor of xG as there is a lack of high-xG shots far away or at tight-angles to the goal, if

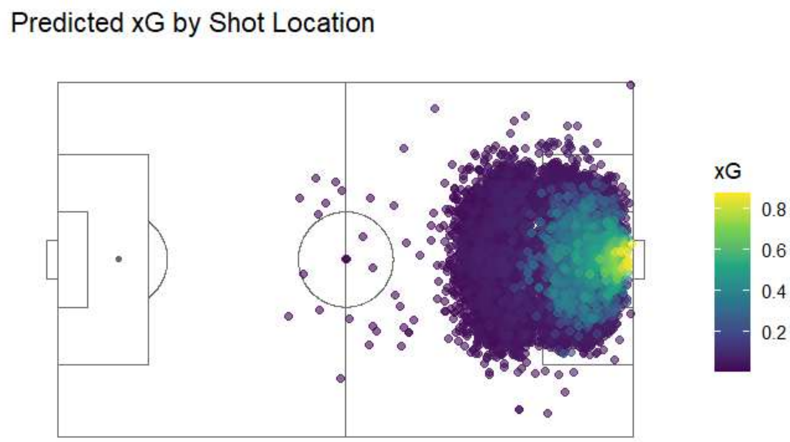

Figure 4: xG Heat Map

other predictors were more influential there would be an expectation that some shots from further away would have a higher xG than shown in the plot.

This section of the results will discuss player over/under performance according to their predicted xG compared to their actual goals scored. The plots are based on a count of the players goals and their xG but only includes players who had at least 20 shots to avoid including players with few shots but good xG performance as it does not truly reflect their true shooting performance due to the lack of data. The plots show the best and worst 10 performers according to their xG respectively.

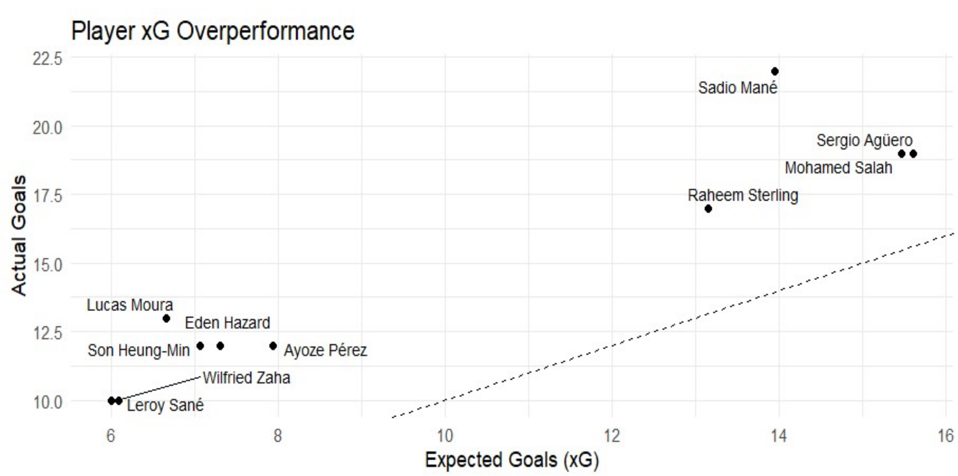

Figure 5: Player xG Overperformance

Combined with Table 4, the plots show that there are a number of players that significantly outperformed their xG this season. The results could suggest that these players outperformed their xG with their skill and footballing ability. However, there must be a consideration that the data used to predict these xGs is missing key positional data. The data did not provide any context about the defending teams position on the pitch at the time of the shot. This data would help produce a more accurate representation of a shot, if there was a number of defenders in between the shooter and the goal, the xG could drop significantly. This would mean that a model that included this tracking data would reward players for their movement and ability to get into better quality shooting positions rather than this model which does not provide that context. In reality, this could change the top over/under performers as currently this model may be subject to a degree of shooting luck. For example, if a defender is blocking the view of the goalkeeper whilst attempting to block a shot, this could increase the xG but in this model it will only ever show that a player is good at shooting.

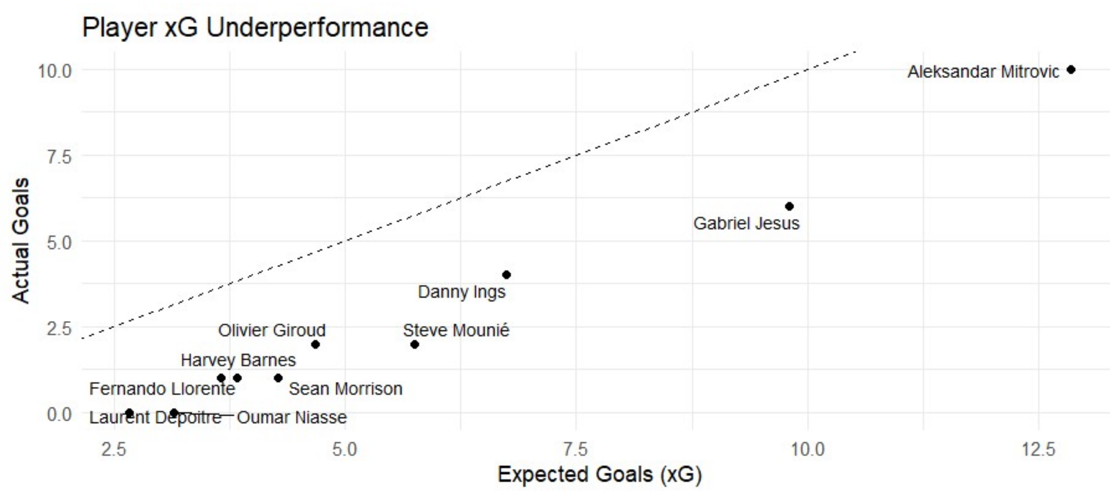

Figure 6: Player xG Underperformance

Table 4 shows the top 5 over and underperformers. Sadio Mané appears to be the best performing player according to xG in this season, this aligns with Liverpool performing well in this season and finishing second.

	wsPlayerName	shots	xg_total	goals	xg_diff
1	Sadio Mané	87	13.95	22	8.05
2	Lucas Moura	49	6.66	13	6.34
3	Son Heung-Min	76	7.07	12	4.93
4	Eden Hazard	89	7.31	12	4.69
5	Ayoze Pérez	55	7.94	12	4.06
6	Gabriel Jesus	43	9.80	6	-3.80
7	Steve Mounié	66	5.75	2	-3.75
8	Sean Morrison	34	4.28	1	-3.28
9	Oumar Niasse	20	3.15	0	-3.15
10	Aleksandar Mitrovic	130	12.85	10	-2.85

Table 4: Player xG Performance

A similar conclusion is found for Moura and Son as Tottenham finished fourth. Gabriel Jesus is the highest xG underperformer, this may be surprising as Manchester City won the league in this season. However, his xG difference means that a lot of his shots were from good positions and he did not convert them. This does align with Manchester City being a strong team as he had a lot of ‘good’ opportunities according to xG but did not convert, so according to the model he should have scored more than he did. Interestingly Mitrovic is the only player in this list to have over 100 shots, far more than anyone else, and despite scoring 10 goals he still underperformed his xG. The combination of a lot of goals and an xG total comparable to the best performer in the league could mean he also did not perform well this season. The underperforming players may not necessarily be the worst performers as the performance summary does take into account overall team strength or defensive position, the overperforming players may have a higher xG as their teammates are providing better passes.

Conclusion

To conclude this report was based on an xG model using one season of premier league data. It tested a variety of models and evaluated their performance based on log loss; it identified that due to its simplicity and performance that GLMNet was the appropriate model to predict xG value. The key predictors identified were distance and angle, which aligned with what was found in the literature, and the key performers aligned with their team’s league position as expected. The limitations of the model such as a lack of tracking data to identify defensive pressure and big chance being a post-shot descriptor means that although the model effectively predicted xG it is not able to meet the predictive capabilities of the top models available, such as Opta or Statsbomb. As there was only one season of data available the predictive capabilities of the model are also lacking, to improve on this more data should be used from a wider range of leagues to properly identify xG performance.

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